

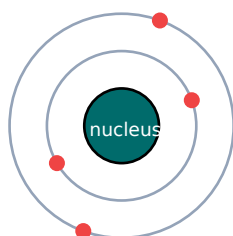
JDScience BTEC Level 3 Structure and Bonding Worksheet

Unit 1 A1 - atomic structure, electron configuration and bonding.

Learning aims

- Interpret atomic number and mass number.
- Explain isotopes and electronic configuration.
- Describe subshells, orbitals and the Aufbau principle at Level 3 standard.
- Explain ionic, covalent and dative covalent bonding.
- Relate bonding to formulae and bond strength.

Atomic structure and the periodic table



Protons: +1

Neutrons: 0

Electrons: -1

Mass number = $p + n$

Atomic number is the number of protons. Mass number is protons plus neutrons. A neutral atom has the same number of protons and electrons.

Isotopes are atoms of the same element with the same number of protons but different numbers of neutrons. Their chemical properties are identical because they have the same electron arrangement.

Electron configuration

At GCSE, electron arrangements are written in shells such as 2,8,1. At BTEC Level 3, shells are divided into subshells labelled s, p, d and f. Electrons fill the lowest energy subshells first, known as the Aufbau principle.

Subshell	Shape / description	Maximum electrons
s	Spherical orbital	2
p	Dumbbell-shaped orbitals	6
d	Five orbitals with varied shapes	10

Ionic bonding

Ionic bonding occurs when electrons are transferred from a metal to a non-metal. The metal forms a positive cation and the non-metal forms a negative anion. Oppositely charged ions are held together by strong electrostatic attraction.

Example: magnesium loses two electrons to form Mg^{2+} . Oxygen gains two electrons to form O^{2-} . The formula is MgO because the charges balance in a 1:1 ratio.

Greater ionic charge gives stronger attraction. Larger ionic radius gives weaker attraction because the ions are further apart.

Covalent and dative bonding

A covalent bond is a shared pair of electrons. It usually forms between non-metals. The shared pair is attracted to both nuclei.

A dative covalent bond is a covalent bond where both electrons in the shared pair come from the same atom, for example when NH_3 donates a lone pair to H^+ .

Student worksheet tasks

A. Atomic structure

Complete a table for K-40, Ca-40, C-12, C-13, S-32 and Al-27 showing protons, neutrons, electrons, atomic number and mass number.

B. Electron configuration

Write shell and subshell configurations for carbon, oxygen, phosphorus, sodium and magnesium.

C. Bonding explanation

Explain, using electron transfer, how magnesium oxide forms. Include charges, formula and why the ionic bond is strong.

D. Dative bonding challenge

Explain how a dative covalent bond differs from an ordinary covalent bond. Use NH_3 and H^+ as an example.